

PREDICTION OF ALPHABET INC. (GOOGL) STOCK PRICE USING CNN, LSTM, AND GRU ALGORITHMS BASED ON K-FOLD CROSS VALIDATION

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Abstract

This study aims to compare the predictive performance of three single deep learning architectures, namely Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU), on the daily closing price of Alphabet Inc. (GOOGL) using an evaluation scheme that preserves the chronological order of the data. The dataset consists of 5,472 daily trading records covering the period from 19 August 2004 to 19 May 2026, obtained from Yahoo Finance. The six price and volume columns were transformed into log-returns, arranged into sliding windows of fifteen days, and scaled using Min-Max normalisation fitted exclusively on the training portion of every fold in order to prevent data leakage. All hyperparameters were determined through seven sequential experiments so that no configuration value appears without an empirical basis. Model evaluation was carried out with a ten-fold Time Series Split cross validation and four complementary metrics, namely Mean Absolute Error, Coefficient of Determination, Directional Accuracy, and Mean Absolute Scaled Error. The results show that none of the three architectures is superior on all four metrics simultaneously, so the best model was determined through a composite ranking. CNN obtained the best composite rank, followed by GRU and LSTM. However, all three models produced a Mean Absolute Scaled Error above one, which means that none of them outperformed the basic forecast. An exact binomial test further shows that the directional accuracy of the three models does not exceed the probability of a coin toss, and the paired effect sizes are negligible. These findings are consistent with the weak form of the efficient market hypothesis and do not support the use of the models as a basis for investment decisions.

Keywords: Stock Price Prediction, Convolutional Neural Network, Long Short-Term Memory, Gated Recurrent Unit, Time Series Cross Validation.

1. INTRODUCTION

Stock prices change every trading day and their direction is difficult to anticipate, so stock price prediction has become a subject that attracts considerable attention in the fields of finance and artificial intelligence. The ability to estimate price movement is useful for understanding the behaviour of the capital market and at the same time serves as a testing ground for various forecasting methods. Alphabet Inc. (GOOGL) was selected as the object of this study because the company has a trading record of more than two decades, its shares are highly liquid, and its historical data are openly available so that the results can be traced and replicated by other researchers. Domestic shares were not selected as the object because this study is a comparison of

architectures rather than a study of one particular market, so the object needs to be a stock that is commonly used in similar research in order for the results to be placed alongside earlier findings. The comparison framework that is tested here can nevertheless be applied to any stock, including those listed on the Indonesia Stock Exchange.

Classical statistical approaches such as ARIMA and exponential smoothing have long been used for financial time series forecasting. These approaches assume a linear relationship between past and future observations, so they tend to be unable to capture the sharp and irregular movements that characterise stock prices. Deep learning offers an alternative because the network learns the relationship pattern directly from the data without the researcher having to specify the

functional form in advance. The disadvantage of this family of models is that it demands a large volume of data, a longer training time, and its results are more difficult to trace.

Three architectures dominate the literature on financial time series forecasting. The Convolutional Neural Network extracts local patterns through a sliding filter, the Long Short-Term Memory maintains long term information through a gated memory cell, and the Gated Recurrent Unit simplifies that structure while retaining comparable capability. Recent work continues that line of attention. Bhanujyothi and Jacob in 2025 proposed a hybrid CNN-LSTM model based on an attention mechanism, and Chohan and colleagues in 2026 developed a hybrid CNN-LSTM-GRU model for a financial risk early warning system. Such work shows that the three architectures are still actively investigated, but they are generally combined into hybrid models and are carried out on markets and contexts that differ from one another.

A systematic review of financial time series forecasting shows that research in this field employs highly varied settings, ranging from the number of layers to the way the data are partitioned, so that the reported results are difficult to compare with one another. A further problem is that many studies still divide the data randomly even though the observations are chronologically ordered, which allows future information to enter the training process. A comparison of single architectures that is carried out fairly on one liquid stock with a validation scheme that preserves the order of time has not been widely examined, so the question of which architecture performs best remains open. Answering that question is the urgency of this study. Accordingly, this study is directed at answering four principal questions, namely which architecture performs best under identical settings, whether the validation scheme produces a stable assessment free of data leakage, which architecture is finally selected together with its error pattern, and what the resulting price projection looks like.

The results of this study are intended for an academic readership, namely researchers and students who work on financial time series forecasting, and not for investors either inside or outside the country. What is offered is an equivalent testing framework together with a candid picture of the limits of the three architectures, so that the findings are used to assess a method and not to make buying or selling decisions.

This study offers an equivalent comparison between CNN, LSTM, and GRU on GOOGL shares. The three models are given identical input, identical

windows, and identical evaluation folds so that any difference in results can be attributed to the architecture rather than to the settings. Assessment employs several complementary error measures so that the conclusion does not depend on a single measure alone.

2. LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

Stock Price and Financial Time Series

A stock price is the value at which a share is traded on an exchange at a given moment. Besides the ordinary closing price, exchanges also record an adjusted closing price. The adjustment accounts for corporate actions such as stock splits and dividend distributions so that the price series remains comparable across time. This study uses the adjusted closing column as the principal target because it is the only column that is free from artificial jumps caused by corporate actions.

A financial time series is a sequence of observations recorded in chronological order. The essential property that distinguishes it from cross sectional data is that the order of observations carries information, so any treatment that breaks the order will damage the pattern that is being learned. A common transformation applied to price series is the log-return, namely the natural logarithm of the ratio between the price on a given day and the price on the preceding day. This transformation converts a series whose level rises over two decades into a series that fluctuates around zero, which is far more suitable as input to a neural network.

Deep Learning and Convolutional Neural Network

Deep learning is a branch of machine learning that arranges processing units into several layers so that the network is able to form representations at successive levels of abstraction. To capture relationships that are not linear, the output of each neuron is passed through an activation function. Without an activation function the addition of layers would not increase the capability of the network at all.

The Convolutional Neural Network was originally developed for image processing but has been adapted to one dimensional data. A filter slides along the time axis and produces a feature map that describes the local pattern within the window. A pooling layer then reduces the size of that map by retaining the largest value in each region, so that the representation becomes more compact and less sensitive to small shifts. The main advantage of this architecture for time

series work is its speed, because the filter operation can be computed in parallel across the entire window.

Long Short-Term Memory and Gated Recurrent Unit

The Long Short-Term Memory was designed to overcome the vanishing gradient problem in ordinary recurrent networks. The architecture maintains a cell state that runs along the sequence and is regulated by three gates, namely the forget gate, the input gate, and the output gate. Each gate is a small network with a sigmoid activation that decides how much information is discarded, added, or released. Through this mechanism the network is able to retain information over relatively long spans.

The Gated Recurrent Unit is a sequential network developed after the Long Short-Term Memory with the aim of simplifying its structure. The underlying idea is the same, namely the use of regulating gates, but the number of gates is reduced to two, namely the update gate and the reset gate, and the cell state is merged with the hidden state. The consequence is a smaller number of parameters at a comparable level of capability, so the Gated Recurrent Unit is often chosen when the volume of data is limited.

Cross Validation That Preserves Chronological Order

Ordinary k-fold cross validation shuffles the observations before partitioning them, which is entirely appropriate for cross sectional data but destructive for time series. If the observations are shuffled, a model may be trained on data from the future and tested on data from the past, a condition known as data leakage. The consequence is that the reported performance becomes optimistic and does not reflect the true ability of the model.

Time Series Split answers that problem by expanding the training window step by step. In the first fold the model is trained on the earliest portion of the data and tested on the segment immediately following it. In the next fold the tested segment is absorbed into the training portion and a new segment is tested, and so on until the final fold. In this way the model is never given information from a period later than the period it is being tested on. This study uses ten folds so that the assessment rests on ten separate test segments rather than on a single partition that might happen to be favourable.

Evaluation Metrics and the Basic Forecast

Mean Absolute Error measures the average magnitude of the deviation between the predicted value and the actual value in the original unit of the

data, so its interpretation is direct. The Coefficient of Determination describes the proportion of the variation in the data that is explained by the model, with a value approaching one indicating a better fit. Directional Accuracy measures the proportion of occasions on which the model correctly anticipates the direction of movement, irrespective of the magnitude of the deviation.

The three measures above do not indicate whether the model is genuinely useful, because a small error may simply be a consequence of a series that does not move much. For that reason this study also uses the Mean Absolute Scaled Error, which compares the error of the model against the error of the basic forecast. The basic forecast, known in the literature as the naive forecast, is the simplest way of forecasting, namely predicting that tomorrow's price will be exactly the same as today's price. A value below one means the model is better than the basic forecast, a value of one means the two are equivalent, and a value above one means the model is worse than simply repeating yesterday's price.

The Efficient Market Hypothesis

The weak form of the efficient market hypothesis states that the price prevailing at present already contains all the information embodied in past prices. If that assumption holds, then studying historical price patterns will not yield a systematic advantage, and the directional accuracy of any model should not differ meaningfully from the probability of a coin toss. This hypothesis provides the theoretical framework for interpreting a directional accuracy that hovers around fifty percent, and it is the reason why this study accompanies the directional figure with an exact binomial test rather than reading the percentage at face value.

Sliding Window and Input Construction

A neural network requires input of a fixed size, whereas a time series is a continuous stream of observations. The sliding window technique bridges that gap by taking a segment of a fixed number of consecutive days as one input sample and the observation on the following day as the target, then shifting the segment forward by one day to form the next sample. The length of the window determines how much history the model is permitted to observe. A window that is too short leaves the model unable to recognise a pattern, whereas a window that is too long enlarges the number of parameters and increases the risk of the model memorising the training data.

This study places all six columns inside the window, so that each sample takes the form of a matrix

of fifteen rows and six columns. The target is a single value, namely the log-return of the adjusted closing column on the day following the window. This arrangement means that the model may make use of the movement of price and volume together, while remaining focused on a single quantity to be predicted.

Scaling, Regularisation, and Optimisation

Min-Max scaling transforms every column into the interval between zero and one by subtracting the minimum value and dividing by the range. The purpose is to place the six columns on a comparable footing, because the volume column has an order of magnitude vastly greater than the price columns and would otherwise dominate the gradient. The critical point in the application of this technique to time series is that the minimum and maximum values must be computed from the training portion alone. If they are computed from the whole dataset, information about the future range enters the training process and the evaluation is no longer honest.

Regularisation is required because a network with tens of thousands of parameters is capable of memorising several thousand training samples, a condition analogous to a student who memorises last year examination papers without understanding the material. Two techniques are applied here. Dropout randomly deactivates a proportion of neurons during training so that the network cannot depend on any single path. The L2 penalty adds the sum of the squared weights to the loss function so that the network is discouraged from producing extreme weights. The strength of both was determined empirically at the second experiment stage.

Optimisation employs the Adam algorithm, which adapts the step size for each parameter on the basis of the first and second moments of the gradient. The learning rate governs how far the weights move at each update. A rate that is too large causes the loss to

oscillate without converging, whereas a rate that is too small makes training slow and prone to becoming trapped in a poor region. The third experiment stage was devoted entirely to determining this value.

Directional Accuracy and Statistical Testing

Directional accuracy is frequently reported in the literature as a headline figure without any accompanying test of significance. That practice is problematic because a value in the vicinity of fifty percent may arise purely by chance. Treating the sequence of directional predictions as a series of independent Bernoulli trials allows the figure to be tested formally against a null probability of one half. The exact binomial test is preferable to a normal approximation when the aim is to state a conclusion with confidence, and the Clopper-Pearson interval provides a companion interval that never falls outside the range between zero and one.

A second question that is often neglected concerns whether the difference between two models is large enough to matter. A significance test answers only whether a difference is distinguishable from zero, not whether it is of any practical consequence. Cohen introduced the standardised effect size for precisely this purpose, with conventional thresholds of 0.2, 0.5, and 0.8 for small, medium, and large effects. Reporting the effect size alongside the significance value therefore protects the reader from over interpreting a difference that is statistically detectable but practically trivial. This study reports both, together with the computed power of the test, so that the limits of the evidence are visible.

Review of Previous Research

A number of studies have compared deep learning architectures on financial time series, with results that vary considerably. A summary of the studies that form the basis of this work is presented in Table 1.

Table 1. Review of Previous Research

Author (Year)	Object	Method	Finding
Fischer & Krauss (2018)	S&P 500 constituents	LSTM vs random forest	LSTM superior, but the split does not preserve time order
Cao, Li, & Li (2019)	Chinese stock indices	CEEMDAN-LSTM hybrid	The hybrid beats a single LSTM
Sezer et al. (2020)	Systematic review	Review of 140 studies	Settings vary widely and are difficult to compare
Joshi et al. (2025)	Comparative review	CNN, LSTM, GRU	No single architecture is consistently superior
Bhanujyothi & Jacob (2025)	Technology shares	Attention CNN-LSTM	Attention improves accuracy on a single market
Chohan et al. (2026)	Financial risk data	CNN-LSTM-GRU hybrid	The hybrid is effective for early warning

Two gaps emerge from the summary. First, most of the studies compare hybrid models rather than single architectures, so the contribution of each architecture cannot be isolated. Second, several of them still partition the data in a manner that does not preserve chronological order, so the reported performance may be optimistic. This study attempts to fill both gaps by comparing three single architectures under identical settings with a validation scheme that preserves the order of time.

Research Questions

Based on the review above, four questions are formulated. First, which of the three architectures obtains the best performance when they are given identical input, identical windows, and identical evaluation folds. Second, whether the ten-fold Time Series Split scheme produces an assessment that is stable across folds and free from data leakage. Third,

which architecture is finally selected on the basis of a composite ranking, and what the pattern of its errors looks like. Fourth, what the shape of the price projection produced by the selected model is when it is run recursively over the following twenty-eight trading days.

3. RESEARCH METHOD

This study is quantitative and experimental in character. The object of the study is the daily share price of Alphabet Inc. (GOOGL), obtained from Yahoo Finance in comma separated value format. The data cover the period from 19 August 2004, the first day the share was listed, to 19 May 2026, amounting to 5,472 trading days. Each record contains a date column and six numerical columns, namely Open, High, Low, Close, Volume, and Adjusted Close. The criteria applied to the research data are presented in Table 2.

Table 2. Research Data Criteria

Description	Total
Daily trading records of GOOGL from 19 August 2004 to 19 May 2026	5,472
Rows lost to the log-return transformation	(1)
Rows lost to the fifteen-day sliding window	(15)
Input samples used in the study	5,456
Test rows per fold (five percent of total rows)	273
Number of evaluation folds	10

The processing stage begins with an examination of data quality. The first step confirms that no row is missing a value in the six columns, because a single empty cell can break a sliding window and render an input sample incomplete. The second step marks trading days whose price movement deviates markedly, using the Tukey interquartile rule with a boundary of one and a half times the interquartile range. This step is not intended to remove data but to document their distribution from the outset. Both steps are carried out before the log-return transformation so

that the condition of the raw data is recorded as it stands.

The six columns are then transformed into log-returns, which reduces the number of usable rows to 5,471 because the first row has no preceding day. The resulting series is arranged into sliding windows of fifteen days, so that each input sample consists of fifteen consecutive days of six columns and the target is the log-return of the adjusted closing column on the sixteenth day. This arrangement yields 5,456 samples. The operational definitions of the variables used in this study are presented in Table 3.

Table 3. Operational Definition of Variables

Variable	Definition	Parameter	Scale
Adjusted Close (Y)	Closing price adjusted for corporate actions, used as the prediction target	$r(t) = \ln(P(t) / P(t-1))$	Ratio
Price Columns (X1)	Open, High, Low, and Close prices of the trading day	Log-return of each column	Ratio
Volume (X2)	Number of shares traded on the trading day	Log-return of volume	Ratio
Window Length (X3)	Number of consecutive days forming one input sample	Sliding window = 15 days	Ratio
Scaling	Range transformation fitted on the training portion only	Min-Max on [0, 1]	Ratio

Scaling employs the Min-Max method with parameters fitted exclusively on the training portion of each fold. The minimum and maximum values of the test portion are never observed at the time of fitting, so no information from a later period can enter the training process. This procedure is the practical guarantee against data leakage in the pipeline of this study.

All configuration values were determined through seven sequential experiment stages so that no number appears without an empirical basis. Experiment E0 determines the length of the sliding window, E1 the capacity of the model, E2 the strength of regularisation, E3 the learning rate, E4 the batch size, E5 the order between partitioning and normalisation, and E6 the weight of the adjusted closing column in the loss function. Every stage was assessed using the Mean Absolute Error on the validation portion, and the winning value was carried forward to the following stage.

Architecture of the Three Models

The three models share the same input and output shape so that the comparison remains equivalent. The input is a matrix of fifteen rows and six columns, and the output is a single value. The CNN model comprises a one dimensional convolution layer with sixteen filters of kernel size three, followed by a max pooling layer, a flattening layer, and a dense layer. The LSTM model comprises two stacked recurrent layers of thirty two units each, followed by a dense layer. The GRU model follows the same arrangement as the LSTM model but with gated recurrent units in place of long short-term memory cells.

The number of trainable parameters was computed analytically and verified against the summary produced by the framework. CNN has 36,166 parameters, LSTM has 70,356, and GRU has 53,331. The figure for LSTM is the largest because each long short-term memory cell maintains four groups of weights, whereas a gated recurrent unit maintains only three and a convolution filter shares its weights across the whole window. The ratio between the number of parameters and the number of training samples ranges from approximately six to thirteen, a level at which regularisation is genuinely necessary.

Loss Function and Training Procedure

The loss function is the mean squared error computed on the log-return of the adjusted closing column. The sixth experiment stage examined whether assigning a larger weight to that column improves the result, and a weight of three was adopted. Training is halted by early stopping with a patience of ten epochs

on the validation loss, and the weights corresponding to the best validation loss are restored. The maximum number of epochs is one hundred, and in practice training halts between forty and seventy epochs.

All computation was carried out in Python with the TensorFlow framework on Google Colab. The random seed was fixed for every library that draws random numbers so that the results can be reproduced. The processing pipeline, from reading the file through to producing the projection, is executed in a single sequence so that there is no manual step that might vary between runs.

Safety Boundary for the Recursive Projection

Recursive projection carries a structural risk. Because the prediction of one day becomes the input for the next, any error committed at the first step is carried into every subsequent step. This behaviour is cumulative, so a small bias may develop into a pronounced trend after several weeks. To prevent the projected trajectory from growing without control, this study imposes a boundary on the log-return that the model is permitted to produce at each step.

The boundary is not chosen arbitrarily but derived from the distribution of the historical log-returns of each column, namely the ninety-nine point nine percentile of their absolute values. For the adjusted closing column this yields a boundary of 0.123487, which corresponds to a daily movement of approximately 13.14 percent. Any predicted log-return exceeding that boundary in absolute value is clipped to the boundary. In practice the boundary was never reached during the twenty-eight step projection, which indicates that the models produce conservative rather than extreme predictions.

To assess whether the resulting trajectory remains plausible, the projected end price is compared against a random walk band. The band is constructed from the daily standard deviation of the log-returns multiplied by the square root of the projection horizon, which for twenty-eight days yields a denominator of 0.101724. A standardised score is then computed from the logarithm of the ratio between the projected end price and the last observed price. A score whose absolute value exceeds 1.96 indicates that the trajectory lies outside the range that a purely random walk would be expected to produce.

Structure of the Evaluation Folds

The structure of the ten evaluation folds is presented in Table 4. Each fold expands the training portion by 273 rows, namely the rows that formed the test portion of the preceding fold, so that the training

portion grows from 2,742 rows in the first fold to 5,199 rows in the tenth.

Fold	Training Rows	Test Rows	Cumulative
1	2,742	273	3,015
2	3,015	273	3,288
3	3,288	273	3,561
4	3,561	273	3,834
5	3,834	273	4,107
6	4,107	273	4,380
7	4,380	273	4,653
8	4,653	273	4,926
9	4,926	273	5,199
10	5,199	273	5,472

The consequence of this structure is that the later folds cover periods in which the price level is far higher than in the early folds. An increase in the absolute error across folds is therefore expected and should not be read as instability of the model. It is for precisely this reason that the study also reports the Mean Absolute Scaled Error, which is invariant to the level of the series.

Model evaluation employs a ten-fold Time Series Split. Five percent of the total rows, namely 273 rows, is allocated to the test portion of each fold, so that ten folds require 2,730 rows and the remaining 2,742 rows form the initial training portion. Each model is retrained from the beginning on every fold so that the assessment is not contaminated by the weights of the preceding fold.

Four evaluation metrics are computed on every fold, namely Mean Absolute Error, Coefficient of Determination, Directional Accuracy, and Mean Absolute Scaled Error. The average across the ten folds is then used as the principal figure, while the standard deviation across folds describes the stability of each model. Because the four metrics do not necessarily point to the same model, the final decision is taken through a composite ranking in which each model is ranked on each metric according to the

direction in which that metric improves, and the four ranks are then averaged.

Directional accuracy that hovers around fifty percent is tested further with an exact binomial test against a null probability of one half, accompanied by a Clopper-Pearson confidence interval. The difference in error between pairs of models across the ten folds is examined with a paired test, a Shapiro-Wilk normality check, a Wilcoxon signed-rank test as a non-parametric companion, and Cohen's effect size so that the practical magnitude of any difference can be assessed rather than merely its statistical significance.

The final stage is a recursive projection over twenty-eight trading days. The selected model predicts the log-return of the following day, that value is used to update the price, the window is shifted forward, and the process repeats. To prevent the accumulated error from causing the projected trajectory to grow without control, a safety boundary is applied in the form of the ninety-nine point nine percentile of the absolute value of the historical log-returns of each column.

4. RESULTS AND DISCUSSION

Descriptive Statistics and Data Quality

The dataset comprises 5,472 trading days across six numerical columns. The results of the descriptive statistical analysis are presented in Table 5.

Table 5. Descriptive Statistics of the Six Columns

Column	Minimum	Maximum	Mean	Std. Deviation
Open	2.48	399.92	60.20	67.46
High	2.55	408.61	60.86	68.28
Low	2.40	396.36	59.55	66.68
Close	2.50	402.62	60.23	67.54
Volume	9,312,000	1,643,023,332	110,524,625	143,623,043
Adjusted Close	2.48	402.38	59.81	67.28

Table 5 shows that the adjusted closing price ranges from 2.48 United States dollars, recorded on 3 September 2004, to 402.38 dollars, recorded on 13

May 2026, with a mean of 59.81 and a standard deviation of 67.28. The standard deviation exceeding the mean indicates an extremely wide distribution, the

natural consequence of a price that grew from approximately two dollars to approximately four hundred dollars over two decades. No empty cell was

found in any of the six columns. The result of the extreme value examination is presented in Table 6.

Column	Lower Bound	Upper Bound	Marked	Percent
Open	-96.8238	197.3462	203	3.71
High	-98.3500	200.2100	206	3.76
Low	-96.1187	195.7912	203	3.71
Close	-96.9113	197.4388	204	3.73
Volume	-120,346,983	280,551,638	517	9.45
Adjusted Close	-96.0713	195.7188	207	3.78

Table 6 shows that the proportion of marked rows in the four price columns lies between 3.71 and 3.78 percent, while the volume column reaches 9.45 percent. None of the marked rows was removed, because removal would break the chronological continuity on which the sliding window depends.

Experiment E0: Sliding Window Length

The first stage examined windows of fifteen, thirty, and sixty days on the three models, producing nine training runs. The results are presented in Table 7.

Table 7. Experiment E0 Results: Sliding Window Length

Model	Window	MAE (USD)	R2	DA (%)	MASE	Param
CNN	15 days	1.9747	0.9894	55.4945	0.9986	9,894
CNN	30 days	2.3924	0.9866	44.3223	1.2099	9,894
CNN	60 days	2.4320	0.9863	44.3223	1.2299	9,894
LSTM	15 days	1.9835	0.9893	52.3810	1.0030	31,906
LSTM	30 days	2.1111	0.9887	43.9560	1.0676	31,906
LSTM	60 days	2.1138	0.9887	43.5897	1.0690	31,906
GRU	15 days	2.0627	0.9888	48.1685	1.0431	24,306
GRU	30 days	2.2052	0.9881	44.8718	1.1152	24,306
GRU	60 days	2.0676	0.9888	43.9560	1.0456	24,306

Table 7 shows that the fifteen day window is superior for all three models. CNN obtains a Mean Absolute Error of 1.9747 dollars with a Mean Absolute Scaled Error of 0.9986, the only value below one across the entire experiment series. Lengthening the window to thirty and sixty days consistently degrades performance, which indicates that the information

relevant to the following day is concentrated in the most recent fortnight.

Experiment E1: Model Capacity

The second stage examined four capacity levels for each model, from small to very large, producing twelve training runs. The results are presented in Table 8.

Table 8. Experiment E1 Results: Model Capacity

Model	Capacity	MAE (USD)	R2	DA (%)	MASE	Param
CNN	Small	2.3044	0.9873	44.1392	1.1653	2,902
CNN	Medium	2.2238	0.9880	45.0549	1.1246	9,894
CNN	Large	2.2477	0.9878	45.4212	1.1366	20,982
CNN	Very large	2.0966	0.9888	45.4212	1.0603	36,166
LSTM	Small	2.1754	0.9882	43.9560	1.1001	8,456
LSTM	Medium	2.0914	0.9888	44.6886	1.0576	31,906
LSTM	Large	2.0136	0.9889	53.6630	1.0183	70,356
LSTM	Very large	2.0421	0.9891	46.3370	1.0327	123,806
GRU	Small	2.1755	0.9883	43.7729	1.1001	6,531
GRU	Medium	2.0627	0.9888	48.1685	1.0431	24,306
GRU	Large	2.0493	0.9885	53.1136	1.0363	53,331
GRU	Very large	2.0608	0.9889	44.6886	1.0421	93,606

Table 8 shows that increasing capacity does not produce a monotonic improvement. CNN attains its best result at the very large level with 36,166 parameters, whereas LSTM and GRU attain theirs at the large level with 70,356 and 53,331 parameters respectively, and both deteriorate when enlarged

further. This pattern is the expected signature of a network beginning to memorise the training data.

Experiment E2: Dropout and L2 Regularisation

The third stage examined four combinations of dropout rate and L2 penalty. The results are presented in Table 9.

Table 9. Experiment E2 Results: Dropout and L2 Regularisation

Model	Dropout / L2	MAE (USD)	R2	DA (%)	MASE
CNN	0.1 and 1e-5	2.0367	0.9891	48.5348	1.0300
CNN	0.2 and 1e-4	2.0736	0.9890	47.9853	1.0486
CNN	0.3 and 1e-4	2.0966	0.9888	45.4212	1.0603
CNN	0.4 and 1e-3	2.0738	0.9889	47.6190	1.0487
LSTM	0.1 and 1e-5	2.0153	0.9889	55.3114	1.0191
LSTM	0.2 and 1e-4	2.0136	0.9889	53.6630	1.0183
LSTM	0.3 and 1e-4	2.1649	0.9883	44.1392	1.0948
LSTM	0.4 and 1e-3	2.2644	0.9876	44.1392	1.1451
GRU	0.1 and 1e-5	2.0562	0.9889	49.0842	1.0398
GRU	0.2 and 1e-4	2.0493	0.9885	53.1136	1.0363
GRU	0.3 and 1e-4	2.2492	0.9877	43.9560	1.1374
GRU	0.4 and 1e-3	2.2965	0.9874	44.6886	1.1613

Table 9 shows that light regularisation is superior for all three models. CNN performs best at a dropout of 0.1 with an L2 penalty of 1e-5, while LSTM and GRU perform best at a dropout of 0.2 with a penalty of 1e-4. Heavier settings degrade performance sharply,

with the Mean Absolute Error of GRU rising from 2.0493 to 2.2965 dollars at the heaviest setting.

Experiment E3: Learning Rate

The fourth stage examined four learning rates spanning two orders of magnitude. The results are presented in Table 10.

Table 10. Experiment E3 Results: Learning Rate

Model	LR	MAE (USD)	R2	DA (%)	MASE
CNN	0.0001	2.0625	0.9890	48.7179	1.0430
CNN	0.0005	2.0654	0.9890	47.8022	1.0445
CNN	0.001	2.0367	0.9891	48.5348	1.0300
CNN	0.005	2.0571	0.9891	45.4212	1.0403
LSTM	0.0001	2.1474	0.9883	44.5055	1.0859
LSTM	0.0005	2.0912	0.9887	45.0549	1.0575
LSTM	0.001	2.0136	0.9889	53.6630	1.0183
LSTM	0.005	2.0017	0.9889	55.6777	1.0123
GRU	0.0001	2.3225	0.9872	44.1392	1.1745
GRU	0.0005	2.0539	0.9888	49.6337	1.0387
GRU	0.001	2.0493	0.9885	53.1136	1.0363
GRU	0.005	2.0516	0.9891	44.5055	1.0375

Table 10 shows that CNN and GRU perform best at a rate of 0.001, whereas LSTM performs best at 0.005 with a Mean Absolute Error of 2.0017 dollars. The lowest rate of 0.0001 is the worst for all three,

which indicates that the training budget of one hundred epochs is insufficient for convergence at that step size.

Experiment E4: Batch Size

The fifth stage examined four batch sizes. The results are presented in Table 11.

Table 11. Experiment E4 Results: Batch Size

Model	Batch	MAE (USD)	R2	DA (%)	MASE
CNN	16	2.0221	0.9892	47.9853	1.0226
CNN	32	2.0367	0.9891	48.5348	1.0300
CNN	64	2.1225	0.9878	54.3956	1.0733
CNN	128	2.1281	0.9878	54.7619	1.0762
LSTM	16	2.0519	0.9882	55.4945	1.0377
LSTM	32	2.0017	0.9889	55.6777	1.0123
LSTM	64	2.3526	0.9870	44.3223	1.1897
LSTM	128	2.0897	0.9879	55.6777	1.0568
GRU	16	2.0556	0.9890	42.6740	1.0395
GRU	32	2.0493	0.9885	53.1136	1.0363
GRU	64	2.0894	0.9887	44.5055	1.0566
GRU	128	2.1177	0.9885	43.9560	1.0709

Table 11 shows that CNN performs best at a batch size of sixteen while LSTM and GRU perform best at thirty two. A batch size of thirty two was adopted uniformly by majority vote, a decision that costs CNN only 0.0146 dollars of Mean Absolute Error and preserves comparability between the three models.

Experiment E5: Order of Normalisation and Partitioning

The sixth stage examined whether the data should be normalised before or after partitioning. The results are presented in Table 12.

Table 12. Experiment E5 Results: Order of Normalisation and Partitioning

Model	Order of steps	MAE (USD)	R2	DA (%)	MASE
CNN	Normalise then split	2.0367	0.9891	48.5348	1.0300
CNN	Split then normalise	2.0367	0.9891	48.5348	1.0300
LSTM	Normalise then split	2.0017	0.9889	55.6777	1.0123
LSTM	Split then normalise	2.0017	0.9889	55.6777	1.0123
GRU	Normalise then split	2.0493	0.9885	53.1136	1.0363
GRU	Split then normalise	2.0493	0.9885	53.1136	1.0363

Table 12 shows that the two orderings produce identical figures to four decimal places for all three models. This result is informative in itself, because it demonstrates that the advantage obtained by normalising before partitioning is negligible on this dataset. The arrangement in which the data are partitioned first was nevertheless adopted, because it is

the only arrangement that guarantees the absence of data leakage.

Experiment E6: Weight of the Adjusted Close Column

The seventh stage examined three weights for the adjusted closing column in the loss function. The results are presented in Table 13.

Table 13. Experiment E6 Results: Weight of the Adjusted Close Column

Model	Weight	MAE (USD)	R2	DA (%)	MASE
CNN	1 (equal)	2.0367	0.9891	48.5348	1.0300
CNN	3 (emphasised)	2.0292	0.9890	51.6484	1.0262
CNN	6 (strongly emphasised)	2.0231	0.9890	52.0147	1.0231
LSTM	1 (equal)	2.0017	0.9889	55.6777	1.0123
LSTM	3 (emphasised)	1.9932	0.9890	55.1282	1.0079
LSTM	6 (strongly emphasised)	1.9978	0.9893	50.0000	1.0103
GRU	1 (equal)	2.0493	0.9885	53.1136	1.0363
GRU	3 (emphasised)	2.0475	0.9885	52.5641	1.0354
GRU	6 (strongly emphasised)	2.0557	0.9888	50.5495	1.0395

Table 13 shows that CNN improves as the weight rises to six, while LSTM and GRU attain their best result at a weight of three. A weight of three was

adopted by majority vote. The recapitulation of all seven stages is presented in Table 14.

Table 14. Recapitulation of the Seven Experiment Stages

Code	Item tested	Cand.	Runs	Best MAE	Worst MAE	Range	Effect
E0	Sliding window length	3	9	1.9747	2.4320	0.4573	0.2434
E1	Model capacity	4	12	2.0136	2.3044	0.2908	0.1652
E2	Dropout / L2	4	12	2.0136	2.2965	0.2829	0.1859
E3	LR	4	12	2.0017	2.3225	0.3208	0.1492
E4	Batch size	4	12	2.0017	2.3526	0.3509	0.1751
E5	Normalisation order	2	6	2.0017	2.0493	0.0476	0.0000
E6	Adjusted Close weight	3	9	1.9932	2.0557	0.0625	0.0101

Table 14 shows that the first stage yields by far the largest effect, with a range of 0.4573 dollars across all rows and an average genuine effect of 0.2434 dollars per model. The fifth stage produces no effect at all, and the seventh produces only 0.0101 dollars. The ordering of the stages therefore matters: the decisions with the greatest consequence are settled first.

Final Configuration and Model Size

The configuration values actually used in all subsequent stages are presented in Table 15, together with the experiment stage from which each value originates.

Table 15. Final Configuration of the Three Models

Item	CNN	LSTM	GRU	Source
Window length (days)	15	15	15	E0
Filters or units	128 and 64	75 and 75	75 and 75	E1
Dense layer units	128	-	-	E1
Dropout	0.1	0.2	0.2	E2
L2 penalty	1e-5	Default	Default	E2
LR	0.001	0.005	0.001	E3
Batch size	32	32	32	E4
Adjusted Close weight	3.0	3.0	3.0	E6
Output layer	Dense(6)	Dense(6)	Dense(6)	Design
Maximum epochs	100	100	100	Fixed
Patience	10	10	10	Fixed
Optimiser	Adam	Adam	Adam	Fixed

The number of trainable parameters was computed analytically and verified against the

summary produced by the framework, as presented in Table 16.

Table 16. Trainable Parameters of the Three Models

Model	Manual count	Framework count	Status	Params per sample
CNN	36,166	36,166	Match	6.63
LSTM	70,356	70,356	Match	12.89
GRU	53,331	53,331	Match	9.77

Table 16 shows that the two computations agree exactly for all three models. The ratio between the number of parameters and the number of training samples ranges from 6.63 for CNN to 12.89 for LSTM, a level at which regularisation is genuinely necessary.

Fold by Fold Evaluation

The four metrics obtained on every fold are presented in Table 17 for CNN, Table 18 for LSTM, and Table 19 for GRU, together with the number of epochs and the training time of each fold.

Table 17. Fold by Fold Results of the CNN Model

Fold	MAE (USD)	R2	DA (%)	MASE	Epoch	Sec.
1	0.4526	0.9153	46.3235	1.0922	10	6.91
2	0.3086	0.9852	50.3676	0.9987	100	65.49
3	0.6205	0.9603	43.3824	1.0978	10	8.61
4	0.7437	0.9113	44.4853	1.0942	10	9.08

Fold	MAE (USD)	R2	DA (%)	MASE	Epoch	Sec.
5	1.1424	0.9614	46.3235	1.0411	10	9.19
6	1.5577	0.9891	43.7500	1.1942	10	9.76
7	2.1932	0.9682	47.7941	1.0210	10	9.16
8	1.6842	0.9781	45.5882	1.0678	10	10.61
9	2.4987	0.9383	47.0588	1.0436	10	10.63
10	3.7127	0.9941	52.9412	1.0459	10	11.55

Table 18. Fold by Fold Results of the LSTM Model

Fold	MAE (USD)	R2	DA (%)	MASE	Epoch	Sec.
1	0.4861	0.9105	44.8529	1.1733	10	8.31
2	0.3113	0.9847	51.8382	1.0075	81	63.64
3	0.8904	0.9349	47.0588	1.5753	10	10.27
4	0.6952	0.9166	50.3676	1.0229	10	11.09
5	1.0997	0.9622	55.8824	1.0022	10	11.21
6	1.3882	0.9907	44.1176	1.0643	10	12.90
7	2.1531	0.9692	51.4706	1.0024	10	13.37
8	1.7200	0.9775	44.4853	1.0906	10	13.06
9	2.6076	0.9352	44.1176	1.0890	10	16.15
10	3.5634	0.9945	55.1471	1.0039	10	14.14

Table 19. Fold by Fold Results of the GRU Model

Fold	MAE (USD)	R2	DA (%)	MASE	Epoch	Sec.
1	0.5246	0.9006	44.4853	1.2661	10	8.33
2	0.3731	0.9798	52.9412	1.2073	10	9.24
3	0.6749	0.9556	45.9559	1.1940	10	10.21
4	0.7724	0.9054	44.1176	1.1365	10	10.42
5	1.1799	0.9591	44.8529	1.0753	10	10.87
6	1.4044	0.9905	45.5882	1.0767	10	11.63
7	2.2161	0.9673	50.3676	1.0317	10	12.94
8	1.6353	0.9784	51.4706	1.0369	10	12.07
9	2.4797	0.9377	45.2206	1.0356	10	13.24
10	3.7190	0.9943	53.6765	1.0477	10	14.33

The three tables show that the Mean Absolute Error rises consistently from approximately 0.45 dollars in the first fold to approximately 3.7 dollars in the tenth. This increase mirrors the rise in the price level itself and is not an indication that the model deteriorates, which is precisely why the Mean Absolute Scaled Error is also reported. The second

fold is notable because it is the only fold in which early stopping did not halt training at ten epochs, and it is also the only fold in which CNN records a Mean Absolute Scaled Error below one, namely 0.9987.

The averages of the four metrics across the ten folds are presented in Table 20.

Table 20. Average Evaluation Metrics across Ten Folds

Model	MAE	Std. Dev.	R Squared	DA (%)	MASE
CNN	1.4914	1.0750	0.9601	46.8015	1.0697
LSTM	1.4915	1.0323	0.9576	48.9338	1.1031
GRU	1.4979	1.0519	0.9569	47.8676	1.1108

Table 20 demonstrates that no single model is superior on all four metrics simultaneously. CNN records the lowest Mean Absolute Error, the highest

Coefficient of Determination, and the smallest Mean Absolute Scaled Error, while LSTM obtains the highest Directional Accuracy at 48.9338 percent and

the smallest standard deviation across folds. Because

the metrics do not converge, the final decision is taken through the composite ranking presented in Table 21.

Table 21. Composite Ranking of the Three Models

Model	Rank MAE	Rank R2	Rank DA	Rank MASE	Average
CNN	1	1	3	1	1.50
LSTM	2	2	1	2	1.75
GRU	3	3	2	3	2.75

On the basis of the composite ranking, CNN is selected as the best model with an average rank of 1.50, followed by LSTM at 1.75 and GRU at 2.75. A far more consequential finding is that all three models record a Mean Absolute Scaled Error above one. The value of 1.0697 for CNN means that the average error of the model is approximately seven percent larger than the average error of the basic forecast, and the values of 1.1031 for LSTM and 1.1108 for GRU are

worse still. None of the three architectures succeeds in outperforming the strategy of simply repeating the previous day's price.

Statistical Testing of Directional Accuracy and Stability

The directional accuracy figures were examined with an exact binomial test across all 2,720 test pairs. The results are presented in Table 22.

Table 22. Exact Binomial Test of Directional Accuracy

Model	Correct	Pairs	DA (%)	z	Sig.	95% CI (%)
CNN	1,273	2,720	46.8015	-3.3363	0.0009	44.91 - 48.70
LSTM	1,331	2,720	48.9338	-1.1121	0.2744	47.04 - 50.83
GRU	1,302	2,720	47.8676	-2.2242	0.0274	45.98 - 49.76

Table 22 shows that the directional accuracy of CNN, at 46.8015 percent, is significantly below fifty percent with an exact probability of 0.0009, and that of GRU, at 47.8676 percent, is likewise significantly below with a probability of 0.0274. LSTM, at 48.9338 percent, does not differ significantly from fifty percent.

None of the three confidence intervals lies above fifty percent.

The stability of each model was examined by comparing the error on the training and test portions on the log-return scale, as presented in Table 23.

Table 23. Model Stability on the Log-Return Scale

Model	Train MAE	Test MAE	Difference	Std. Dev.	Cohen's d
CNN	0.044901	0.044653	-0.000248	0.010932	-0.0227
LSTM	0.045432	0.045769	+0.000337	0.010548	+0.0320
GRU	0.045958	0.046067	+0.000108	0.010469	+0.0103

Table 23 shows that the difference between training and test error is below 0.0004 in absolute value for all three models, with effect sizes of negative 0.0227, positive 0.0320, and positive 0.0103. All three

are conventionally regarded as negligible, which indicates that none of the models is overfitting. Four supporting tests were applied to confirm this reading, as presented in Table 24.

Table 24. Four Supporting Tests of Model Stability

Model	Shapiro W	p	t	p	Wilcoxon W	p	Power
CNN	0.9480	0.6450	-0.0716	0.9445	25.0	0.8457	0.0505
LSTM	0.9437	0.5952	+0.1011	0.9217	27.0	1.0000	0.0509
GRU	0.9606	0.7929	+0.0327	0.9747	27.0	1.0000	0.0501

Table 24 shows that the Shapiro-Wilk test confirms normality for all three models, the paired t test yields probabilities between 0.9217 and 0.9747, and the Wilcoxon signed-rank test agrees. The computed power ranges from 0.0501 to 0.0509, which means that with only ten folds the study is not in a

position to detect a difference of this magnitude even if one exists.

Prediction Results and Recursive Projection

The final stage produced a recursive projection over twenty-eight trading days beyond the last observed trading day, namely 19 May 2026 at a

closing price of 387.43 dollars. The summary of the three projected trajectories is presented in Table 25.

Table 25. Summary of the Twenty-Eight Day Projected Trajectories

Model	Period	Days	Low	High	Start	End	Change	%
CNN	May 2026	7	382.91	387.58	387.51	382.91	-4.61	-1.19
CNN	Jun 2026	21	296.34	381.37	381.37	296.34	-85.03	-22.30
LSTM	May 2026	7	371.27	385.30	385.30	371.27	-14.04	-3.64
LSTM	Jun 2026	21	322.39	368.86	368.86	322.39	-46.47	-12.60
GRU	May 2026	7	386.24	387.26	387.26	386.24	-1.02	-0.26
GRU	Jun 2026	21	385.35	386.21	386.21	385.35	-0.87	-0.22

Table 25 shows that the three trajectories differ markedly in character. The CNN trajectory declines gently through the seven remaining trading days of May, from 387.51 to 382.91 dollars, a fall of 1.19 percent, and then falls sharply through the twenty-one trading days of June, from 381.37 to 296.34 dollars, a fall of 22.30 percent. The LSTM trajectory falls by 3.64 percent in May and 12.60 percent in June, ending

at 322.39 dollars. The GRU trajectory remains almost flat throughout, falling by only 0.26 percent in May and 0.22 percent in June, ending at 385.35 dollars.

To assess whether these trajectories remain plausible, the projected price on the twenty-eighth day was compared against a random walk band constructed from the daily standard deviation of the log-returns. The results are presented in Table 26.

Table 26. Position of the Day Twenty-Eight Price against the Random Walk Band

Model	Projected price (USD)	z score	Position
CNN	296.34	-2.63	Outside the 95% band
LSTM	322.39	-1.81	Inside the 95% band
GRU	385.35	-0.05	Inside the 95% band

Table 26 shows that the CNN projection yields a standardised score of negative 2.63 and therefore falls outside the ninety-five percent band, whereas the LSTM and GRU projections remain inside it with scores of negative 1.81 and negative 0.05. The CNN trajectory, which belongs to the model selected by the composite ranking, is thus the least plausible of the

three when judged against what a purely random walk would produce.

A diagnostic comparison was carried out between the earlier design, in which five of the six columns were frozen during projection, and the final design in which no column is frozen. The results are presented in Table 27.

Table 27. Distribution of the Projected Log-Returns under Two Designs

Model	Design	Mean	Std. Dev.	Range	Ratio
CNN	Old, five columns frozen	-0.009091	0.004514	0.013130	0.2348
CNN	New, no freezing	-0.009572	0.005850	0.018371	0.3043
LSTM	Old, five columns frozen	-0.006750	0.000300	0.001464	0.0156
LSTM	New, no freezing	-0.006563	0.000324	0.001296	0.0168
GRU	Old, five columns frozen	-0.003187	0.000698	0.003078	0.0363
GRU	New, no freezing	-0.000193	0.000178	0.000654	0.0093

Table 27 shows that removing the freezing raises the ratio of the projected log-return volatility to the historical volatility for CNN from 0.2348 to 0.3043, while for GRU it falls from 0.0363 to 0.0093. Neither design produces a projection whose volatility approaches the historical level, which confirms that the flat character of the trajectories is a property of the

models themselves rather than an artefact of the earlier code defect.

Comparison with Previous Findings

A comparison between the present results and three recent Indonesian studies on the same family of architectures is presented in Table 28.

Table 28. Comparison with Previous Research

No	Researcher and Year	Object	Algorithm	Reported measure	Value
1	Aryati et al. (2024)	UNVR (IDX)	LSTM and GRU	MAPE and RMSE	GRU: MAPE 2.14%, RMSE 0.01775
2	Fitriani et al. (2025)	ASII and ADRO (IDX)	LSTM and GRU	RMSE and MAPE	GRU (ASII): RMSE 80.411, MAPE 1.329
3	Anzari et al. (2025)	BBRI (IDX)	LSTM and GRU	R2 and MAPE	GRU better: R2 +10.7%, MAPE -18.5%
4	This study (2026)	GOOGL (NASDAQ)	CNN, LSTM, GRU	MASE and DA	CNN: MASE 1.0697, DA 46.80%

Table 28 shows that the three earlier studies report percentage or absolute error measures that appear impressive in isolation, such as a Mean Absolute Percentage Error of 2.14 percent and a Root Mean Squared Error of 0.01775. None of them reports a scaled measure against the basic forecast, and none reports a formal test of directional accuracy. The present study reports both, and it is precisely those two additions that reveal a result which the conventional measures conceal. The discrepancy is therefore not evidence of a weaker implementation but of a stricter protocol.

A second source of discrepancy lies in the partitioning of the data. Studies that shuffle observations before splitting allow information from later periods to inform the training process, and the reported performance is consequently optimistic. The present study deliberately avoids that practice.

Implications for Theory and Practice

For theory, the results provide empirical support for the weak form of the efficient market hypothesis on a highly liquid share over a period of more than two decades. If historical price patterns carried systematically exploitable information, at least one of the three architectures would be expected to exceed the probability of a coin toss in anticipating direction. None of them does, and two of them fall significantly below it.

For practice, the implication is a caution rather than a recommendation. A model that produces a small numerical error is not necessarily useful, because the relevant comparison is not against zero error but against the simplest available alternative. Any study reporting the performance of a forecasting model on financial data should therefore accompany the conventional error measures with a scaled measure and with a formal test of directional accuracy.

The narrowness of the margin between the three architectures further suggests that effort invested in selecting between CNN, LSTM, and GRU is likely to yield less benefit than effort invested in enriching the input. The three models were given identical

information and arrived at results that cannot be distinguished statistically, which is itself evidence that the limiting factor lies in the information contained in price and volume history rather than in the capacity of the architecture to extract it.

5. CONCLUSION, LIMITATION, AND SUGGESTION

Conclusion

This study compared three single deep learning architectures on the daily share price of Alphabet Inc. using an evaluation scheme that preserves chronological order. On the basis of the composite ranking, CNN obtains the best position with an average rank of 2.00, followed by GRU at 2.25 and LSTM at 2.50. The ten-fold Time Series Split scheme succeeds in preventing data leakage, although the absolute error varies widely across folds because the price level rises over time. The selected model displays an error pattern that is comparatively accurate in magnitude but poor in direction. The recursive projection of the selected model falls outside the random walk band, so it should be treated as an illustration rather than as an estimate. Above all, none of the three models outperforms the basic forecast, and the directional accuracy of the three does not exceed the probability of a coin toss, so the findings do not support the use of these models as a basis for investment decisions.

Limitation

This study is confined to a single share, so the findings cannot be generalised to other shares or to other markets without further testing. The input is restricted to price and volume columns, without macroeconomic variables, news sentiment, or technical indicators. The evaluation employs ten folds, a number that yields a test power of only 0.0505 for the effect sizes encountered, so the study is not in a position to detect small differences between architectures. The recursive projection is also limited to twenty-eight trading days because the accumulation of error renders any longer horizon uninformative.

Suggestion

Subsequent researchers are encouraged to extend the object of study to several shares across different sectors so that the stability of the comparison can be assessed. The addition of exogenous variables such as market indices, exchange rates, or news sentiment is worth examining, because the present findings indicate that price and volume columns alone are insufficient. Increasing the number of folds or employing a blocked bootstrap procedure would raise the power of the test. Finally, future work should report the Mean Absolute Scaled Error alongside the conventional error measures, because that measure is what reveals whether a model genuinely offers anything beyond the basic forecast.

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